

International multidimensional authenticity specification (IMAS) algorithm for detection of commercial pomegranate juice adulteration.

The pomegranate fruit (*Punica granatum*) has become an international high-value crop for the production of commercial pomegranate juice (PJ). The perceived consumer value of PJ is due in large part to its potential health benefits based on a significant body of medical research conducted with authentic PJ. To establish criteria for authenticating PJ, a new International Multidimensional Authenticity Specifications (IMAS) algorithm was developed through consideration of existing databases and comprehensive chemical characterization of 45 commercial juice samples from 23 different manufacturers in the United States. In addition to analysis of commercial juice samples obtained in the United States, data from other analyses of pomegranate juice and fruits including samples from Iran, Turkey, Azerbaijan, Syria, India, and China were considered in developing this protocol. There is universal agreement that the presence of a highly constant group of six anthocyanins together with punicalagins characterizes polyphenols in PJ. At a total sugar concentration of 16 degrees Brix, PJ contains characteristic sugars including mannitol at >0.3 g/100 mL. Ratios of glucose to mannitol of 4-15 and of glucose to fructose of 0.8-1.0 are also characteristic of PJ. In addition, no sucrose should be present because of isomerase activity during commercial processing. Stable isotope ratio mass spectrometry as $\delta^{13}C$ > -25 per thousand assures that there is no added corn or cane sugar added to PJ. Sorbitol was present at <0.025 g/100 mL; maltose and tartaric acid were not detected. The presence of the amino acid proline at >25 mg/L is indicative of added grape products. Malic acid at >0.1 g/100 mL indicates adulteration with apple, pear, grape, cherry, plum, or aronia juice. Other adulteration methods include the addition of highly concentrated aronia, blueberry, or blackberry juices or natural grape pigments to poor-quality juices to imitate the color of pomegranate juice, which results in abnormal anthocyanin profiles. To adjust the astringent taste of poor-quality juice or peel extract, addition of nonpomegranate sugars is a commonly detected adulteration method. The profile generated from these analyses combined with information from existing databases and published literature has been integrated into a validated IMAS for PJ, which can be utilized to detect PJ adulteration. In this survey of commercial pomegranate juices, only 6 of 23 strictly met all of the IMAS criteria.